

Able Baker Problem

A computer technical support center is staffed by two people – Able and Baker who take calls and try to answer questions and solve computer problems. The time between calls ranges from 1 to 4 minutes, with distribution as shown in the table given below

Inter-arrival time (mins)	Probability
1	0.25
2	0.40
3	0.20
4	0.15

Able is more experienced and can provide service faster than Baker. The distributions of their service times are shown in following tables. When both are idle, Able takes the call. If both are busy, the customer has to wait.

Distribution of Able's Service Time

Service time (mins)	Probability
2	0.30
3	0.28
4	0.25
5	0.17

Distribution of Baker's Service Time

Service time (mins)	Probability
3	0.35
4	0.25
5	0.20
6	0.20

Problem statement

- Simulate the system for 15 customers and compute average inter-arrival time, average service time of Able, average service time of Baker, average waiting time of customers, probability of idle time of Able, probability of idle time of Baker.
- The random numbers for inter-arrival time are: 26, 98, 90, 26, 42, 74, 80, 68, 22, 48, 34, 45, 24, 34.
- The random numbers for service time are: 95, 21, 51, 92, 89, 38, 13, 61, 50, 49, 39, 53, 88, 01, 81.

cumulative probability and range of random numbers are computed for inter-arrival time and service time of both Able and Baker

Inter-arrival Time Table

Inter-arrival time (mins)	Probability	Cumulative Probability	Range of Random Numbers
1	0.25	0.25	0 – 25
2	0.40	0.65	26 – 65
3	0.20	0.85	66 – 85
4	0.15	1.00	86 – 00

Distribution of Able's Service Time

Service time (mins)	Probability	Cumulative Probability	Range of Random Numbers
2	0.30	0.30	0 – 30
3	0.28	0.58	31 – 58
4	0.25	0.83	59 – 83
5	0.17	1.00	84 – 00

Distribution of Baker's Service Time

Service time (mins)	Probability	Cumulative Probability	Range of Random Numbers
3	0.35	0.35	0 – 35
4	0.25	0.60	36 – 60
5	0.20	0.80	61 – 80
6	0.20	1.00	81 – 00

Random numbers for simulation

The random numbers for inter-arrival time are:

26, 98, 90, 26, 42, 74, 80, 68, 22, 48, 34, 45, 24, 34.

The random numbers for service time are:

95, 21, 51, 92, 89, 38, 13, 61, 50, 49, 39, 53, 88, 01,
81.

Simulation Table

Cu st No (1)	Inter- arriva l time (2)	Arriv al time in Clock (3)	Rando m Number for Service Time (4)	Able				Baker				Time in Q (13)
				Service time begins (5)	Service time (6)	Service time ends (7)	Idle tim e (8)	Service time begins (9)	Service time (10)	Service time ends (11)	Idle tim e (12)	
1	-	0	95	0	5	5	0					0
2	2	2	21					2	3	5	2	0
3	4	6	51	6	3	9	1					0
4	4	10	92	10	5	15	1					0
5	2	12	89					12	6	18	7	0
6	2	14	38	15	3	18	0					1
7	3	17	13	18	2	20	0					1
8	3	20	61	20	4	24	0					0
9	3	23	50					23	4	27	5	0
10	1	24	49	24	3	27	0					0
11	2	26	39	27	3	30	0					1
12	2	28	53					28	4	32	1	0
13	2	30	88	30	5	35	0					0
14	1	31	01					32	3	35	0	1
15	2	33	81	35	4	39	0					2
To tal	33				37		2		20		15	6

Simulation table is prepared as instructed below –

(i) Inter-arrival time (Col2) is computed using the given random numbers and “Interarrival Time Table”.

(ii) Time in Clock (Col3) is cumulative sum of Col2.

(iii) Col4 is filled with random numbers given for service time.

(iv) As it is given that, when both Able and Baker are free, Able will serve the customer, Service begins for Able (Col5) is 0. Use the random number (Col4) and compare it with “Distribution of Able’s Service Time” table to get the Col6 value of 1st row.

$\text{Col7} = \text{Col5} + \text{Col6}$

(v) Observe when next customer arrives (Col3). Allot him to either Able or Baker depending on their availability. Here, 2nd customer is assigned to Baker, as Able was busy. Find Col9 accordingly. Col10 is computed using the random number.

$\text{Col11} = \text{Col9} + \text{Col10}$.

(vi) Perform the procedure for all customers.

(vii) Idle time of Able (Col8) = Col5 of next served customer – Col7 of previously served customer. Similarly, Idle time of Baker (Col12) = Col9 of next served customer – Col11 of previously served customer

(viii) Time in Queue (Col13) = Either of Col5 or Col9 – Col3

Performance Measures:

(i) Average Inter-arrival time = $33/14 = 2.36$ mins

(ii) Average service time of Able =

Total service time of Able / Number of people Able served = $37/10 = 3.7$ mins

(iii) Average service time of Baker =

Total service time of Baker / Number of people Baker served = $20/5 = 4$ mins

(iv) Average waiting time of customers =

Total waiting time / Number of customers = $6/15 = 0.4$ mins

(v) Average waiting time of customers who wait =

total waiting time / number of customers who wait = $6/5 = 1.2$ mins

(vi) Probability of Able's idle time =

Able's idle time / total run time of simulation = $2/39 = 0.05$

(vii) Probability of Baker's idle time

= Baker's idle time / total run time of simulation = $15/39 = 0.385$

(viii) Probability that a customer has to wait in a queue =

Number of customers who wait / total number of customers = $5/15 = 0.33$

Simulation Table

Cu st No (1)	Inter- arriva l time (2)	Arriv al time in Clock (3)	Rando m Number for Service Time (4)	Able				Baker				Time in Q (13)
				Service time begins (5)	Service time (6)	Service time ends (7)	Idle tim e (8)	Service time begins (9)	Service time (10)	Service time ends (11)	Idle tim e (12)	
1	-	0	95	0	5	5	0					0
2	2	2	21					2	3	5	2	0
3	4	6	51	6	3	9	1					0
4	4	10	92	10	5	15	1					0
5	2	12	89					12	6	18	7	0
6	2	14	38	15	3	18	0					1
7	3	17	13	18	2	20	0					1
8	3	20	61	20	4	24	0					0
9	3	23	50					23	4	27	5	0
10	1	24	49	24	3	27	0					0
11	2	26	39	27	3	30	0					1
12	2	28	53					28	4	32	1	0
13	2	30	88	30	5	35	0					0
14	1	31	01					32	3	35	0	1
15	2	33	81	35	4	39	0					2
To tal	33				37		2		20		15	6